

Assessment 6.1: Food Systems Quiz

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.1 Where Food Comes From (Lessons 6.1 to 6.5) **When:** the last 15 minutes of Lesson 6.5, Day 2 **Length:** 12 items, 15 minutes **Category:** Topic assessments (25 percent of the course grade) **Points:** 21 **Retakes:** one retake after a review, best score counts (Grading Plan) **Allowed:** the unit vocabulary card. Wall paths and notes are covered.

Food Systems Quiz

Name: _____ Date: _____ Period: _____

Directions: You have 15 minutes. Answer every question. Write in pen or pencil. For the short answer questions, a complete sentence is enough; you do not need a paragraph. You may use your vocabulary card.

Part A. Put it in order (3 points)

1. Here are six steps a loaf of bread takes. Number them 1 to 6 in the order they happen.

____ A truck carries the loaves to a warehouse. ____ A bakery mixes and bakes the dough. ____ A farmer grows wheat. ____ A store puts the loaves on a shelf. ____ A mill grinds the wheat into flour. ____ A family eats the bread.

2. In the chain above, which step usually adds the most to the price of the loaf: growing the wheat, or everything that happens after it? Circle one and give one reason.

Growing the wheat / Everything after it

Reason: _____

Part B. Local and seasonal (4 points)

3. Circle the three foods that grow in New York State.

apples bananas potatoes coffee cabbage rice

4. It is January in New York. Which one of these is the most realistic local choice for dinner? Circle one.

- a) fresh strawberries
- b) a potato from storage
- c) a mango
- d) fresh corn on the cob

5. A tomato bought here in February tastes weaker than one bought in August. Give one reason, in one sentence.



6. Name one advantage and one disadvantage of buying food from a local farm instead of a supermarket.

Advantage: _____

Disadvantage: _____

Part C. Waste math (4 points)

Use these numbers. Show your work.

A class of 24 students collected food scraps for 2 days. The edible bin weighed 1,440 grams in total for the 2 days.

7. How many grams of edible waste is that per student per day? Show your work.

Work: _____

Answer: _____ grams per student per day

8. At that rate, how many grams would one student waste in 180 school days? Show your work.

Work: _____

Answer: _____ grams

9. One pound is about 454 grams. About how many pounds is your answer to question 8? Round to the nearest whole pound.

Answer: about _____ pounds

Part D. A food desert scenario (3 points)

The Alvarez family lives on Pine Street. The nearest full supermarket is 1.4 miles away. There are two convenience stores within three blocks. The bus runs once an hour. Nobody in the household drives.

10. Name two reasons this family may end up buying most of their food at the convenience store, even though it costs more.

Reason 1: _____

Reason 2: _____

Part E. Sustainable methods (4 points)

11. Match each method to what it does. Write the letter on the line.

___ Crop rotation ___ Drip irrigation ___ Integrated pest management ___ Composting

a) Turns food scraps and plant waste into soil

- b) Changes what is planted in a field each year so the soil keeps its nutrients and pests lose their host
- c) Watches for pests first and uses the smallest response that works, spraying last instead of first
- d) Delivers water to the roots instead of spraying it over the whole field, so less evaporates

Part F. Packaging trade-off, short answer (3 points)

12. A cucumber wrapped in plastic stays fresh about three times longer than one sold loose. The wrap is a piece of single use plastic.

A store manager has to pick one. Which would you tell them to sell, and why? Name the trade-off in your answer: what gets better and what gets worse.







End of quiz. Reread question 12 if you have time left.

Teacher key

Total: 21 points. Anything within the described range earns the point. A student who gets the reasoning right and the arithmetic wrong loses the arithmetic point only.

Item	Points	Answer	Standard line this item checks
1	2 (all six correct, or 1 point for four or five correct)	Order: 4 (truck), 3 (bakery), 1 (farmer), 5 (store), 2 (mill), 6 (family)	NYS FSP 3. Current Issues Related to Food in Global Societies a) Describe factors affecting the food supply... (Lesson 6.1)
2	1	“Everything after it.” Accept any reason naming processing, transport, packaging, storage, or the store’s share; accept “the farmer gets the smallest part”	NYS FSP 3. a); National FCS 14.4.3 (Lesson 6.1)
3	1 (all three)	apples, potatoes, cabbage	NYS FSP 2. Food Preparation a) Identify specific foods from the region being studied (Lesson 6.2)
4	1	b) a potato from storage	NYS FSP 2. a); NYS FSP 2. b) (Lesson 6.2)
5	1	Accept: it was picked before it was ripe so it could survive the trip; it traveled a long way and sat in storage; it was grown out of season in a greenhouse or far away. Do not require the word “ripe”	NYS FSP 2. Food Preparation b) Explore factors affecting regional food choices; National FCS 14.1.4 (Lesson 6.2)

6	1 (both sides needed)	Advantage: fresher, better taste, shorter trip, money stays local, supports local jobs. Disadvantage: costs more, fewer choices, not available in winter, store hours or distance	NYS FSP 4. Future of Food in a Global Society b) Explore the advantages and disadvantages of selecting and purchasing sustainable food products; CTE SUS 2. h) (Lesson 6.2)
7	2 (1 for the work, 1 for the answer)	1,440 divided by 24 equals 60 grams for 2 days; 60 divided by 2 equals 30 grams per student per day	NYS FSP 3. a); CDOS 3a Basic Skills (Lesson 6.3)
8	1	30 times 180 equals 5,400 grams	Same as item 7 (Lesson 6.3)
9	1	5,400 divided by 454 equals about 12 pounds (accept 11 to 12)	Same as item 7 (Lesson 6.3)
10	3 (1 for each reason, plus 1 if at least one reason names distance or transportation rather than only price)	Accept any two of: 1.4 miles is too far to carry groceries on foot; nobody drives; the bus runs only once an hour so a trip takes most of a day; frozen or refrigerated food will not survive the wait; the convenience store is three blocks away and open late; a small trip now is easier than a big trip later	NYS FSP 4. Future of Food in a Global Society d) Identify the consequences of food scarcity and hunger on society; NYS FSP 5. c); CTE FCL 4. g) (Lesson 6.4)
11	4 (1 each)	Crop rotation b; Drip irrigation d; Integrated pest management c; Composting a	NYS FSP 4. Future of Food in a Global Society a) Understand the components of

			sustainable food systems; CTE SUS 1. a); National FCS 2.2.3 (Lesson 6.5)
12	3 (1 for a clear choice, 1 for naming what gets better, 1 for naming what gets worse)	Either choice earns full credit if the trade-off is named on both sides. Wrapped: less food waste and longer shelf life, but more plastic in the trash. Loose: less plastic, but more cucumbers spoil and get thrown out. Full credit for a middle answer (“sell both and let the customer pick”) if both sides are still named	NYS FSP 4. Future of Food in a Global Society e) Explore technological advances that affect the global food supply; CTE SUS 2. h); National FCS 8.5.13 (Lesson 6.5)

Scoring bands

Points	Meaning
19 to 21	Meets and exceeds. Ready for Topic 6.2 with no review
15 to 18	Meets. Reteach whichever part lost points, in two minutes, at the start of Lesson 6.6
11 to 14	Approaching. Retake offered after the review
Below 11	Not yet. Retake required, with a five-minute conference first to find out whether the problem is the content or the reading

How to use

- Item 7 is the one most students lose. The two-step division (by students, then by days) is the whole difficulty. If more than a third of the class misses it, that is a math reteach, not a content reteach, and it takes three minutes on the board.
- Items 6 and 12 are both trade-off items with no single right answer. Score the reasoning, not the position. A student who picks the answer you disagree with and defends it on both sides gets full credit.

- Item 10 is the one to read before you hand the quizzes back. A student answer that reveals something personal about their own household gets a private conversation, not a comment in the margin, and see the three rules in Lesson 6.4.
- Retake version: change the numbers in Part C, swap the six-step chain in item 1 to milk, and keep everything else. The retake is the same test with new numbers, not an easier one.